

18. $r = a$; $\rho(r, \theta) = \frac{1}{1 + r^4}$; I_x
19. Outside $r = a$ and inside $r = 2a \cos \theta$; density at a point P inversely proportional to the cube of the distance from the pole; I_y
20. Outside $r = 1$ and inside $r = 2 \sin 2\theta$, first quadrant; $\rho(r, \theta) = \sec^2 \theta$; I_y

In Problems 21–24, find the **polar moment of inertia**

$$I_0 = \iint_R r^2 \rho(r, \theta) dA = I_x + I_y$$

of the lamina that has the given shape and density.

21. $r = a$; $\rho(r, \theta) = k$ (constant) [Hint: Use Problem 17 and the fact that $I_x = I_y$.]
22. $r = \theta$, $0 \leq \theta \leq \pi$, $y = 0$; density at a point P proportional to the distance from the pole
23. $r\theta = 1$, $\frac{1}{3} \leq \theta \leq 1$, $r = 1$, $r = 3$, $y = 0$; density at a point P inversely proportional to the distance from the pole [Hint: Integrate first with respect to θ .]
24. $r = 2a \cos \theta$; $\rho(r, \theta) = k$ (constant)

In Problems 25–32, evaluate the given iterated integral by changing to polar coordinates.

25. $\int_{-3}^3 \int_0^{\sqrt{9-x^2}} \sqrt{x^2 + y^2} dy dx$
26. $\int_0^{\sqrt{2}/2} \int_0^{\sqrt{1-y^2}} \frac{y^2}{\sqrt{x^2 + y^2}} dx dy$
27. $\int_0^1 \int_0^{\sqrt{1-y^2}} e^{x^2+y^2} dx dy$
28. $\int_{-\sqrt{\pi}}^{\sqrt{\pi}} \int_0^{\sqrt{\pi-x^2}} \sin(x^2 + y^2) dy dx$
29. $\int_0^1 \int_{\sqrt{1-x^2}}^{\sqrt{4-x^2}} \frac{x^2}{x^2 + y^2} dy dx + \int_1^2 \int_0^{\sqrt{4-x^2}} \frac{x^2}{x^2 + y^2} dy dx$
30. $\int_0^1 \int_0^{\sqrt{2y-y^2}} (1 - x^2 - y^2) dx dy$
31. $\int_{-5}^5 \int_0^{\sqrt{25-x^2}} (4x + 3y) dy dx$
32. $\int_0^1 \int_0^{\sqrt{1-y^2}} \frac{1}{1 + \sqrt{x^2 + y^2}} dx dy$

33. The liquid hydrogen tank in the space shuttle has the form of a right circular cylinder with a semi-ellipsoidal cap at each end. The radius of the cylindrical part of the tank is 4.2 m. Find the volume of the tank shown in **FIGURE 9.11.6**.

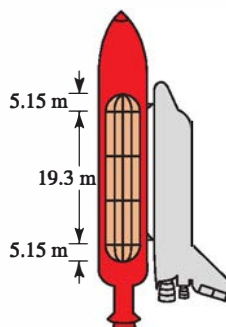


FIGURE 9.11.6 Fuel tank in Problem 33

34. Evaluate $\iint_R (x + y) dA$ over the region shown in **FIGURE 9.11.7**.

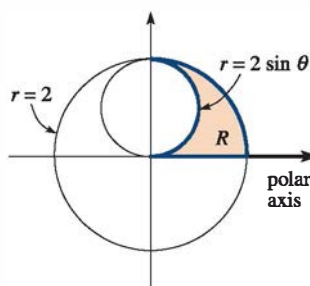


FIGURE 9.11.7 Region R for Problem 34

Discussion Problem

35. The improper integral $\int_0^\infty e^{-x^2} dx$ is important in the theory of probability, statistics, and other areas of applied mathematics. If I denotes the integral, then

$$I = \int_0^\infty e^{-x^2} dx \quad \text{and} \quad I = \int_0^\infty e^{-y^2} dy$$

and consequently

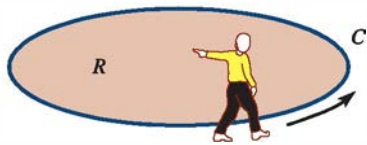
$$I^2 = \left(\int_0^\infty e^{-x^2} dx \right) \left(\int_0^\infty e^{-y^2} dy \right) = \int_0^\infty \int_0^\infty e^{-(x^2+y^2)} dx dy.$$

Discuss how to use polar coordinates to evaluate the last integral. Find the value of I .

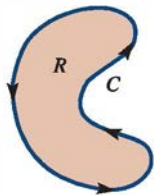
9.12 Green's Theorem

Introduction One of the most important theorems in vector integral calculus relates a line integral around a piecewise-smooth simple closed curve C to a double integral over the region R bounded by the curve.

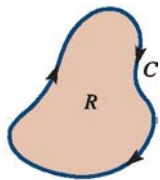
Line Integrals Along Simple Closed Curves We say the **positive direction** around a simple closed curve C is that direction a point on the curve must move, or the direction a person



(a) Positive direction



(b) Positive direction



(c) Negative direction

FIGURE 9.12.1 Directions around curve C

must walk on C , in order to keep the region R bounded by C to the left. See **FIGURE 9.12.1(a)**. Roughly, as shown in Figure 9.12.1(b) and 9.12.1(c), the *positive* and *negative* directions correspond to the *counterclockwise* and *clockwise* directions, respectively. Line integrals on simple closed curves are written

$$\oint_C P(x, y) dx + Q(x, y) dy, \quad \oint_C P(x, y) dx + Q(x, y) dy, \quad \oint_C F(x, y) ds,$$

and so on. The symbols $\oint_C$ and $\oint_C$ refer, in turn, to integrations in the positive and negative directions.

Theorem 9.12.1 Green's Theorem in the Plane

Suppose that C is a piecewise-smooth simple closed curve bounding a simply connected region R . If P , Q , $\partial P/\partial y$, and $\partial Q/\partial x$ are continuous on R , then

$$\oint_C P dx + Q dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA. \quad (1)$$

PARTIAL PROOF: We shall prove (1) only for a region R that is simultaneously of Type I and Type II:

$$R: g_1(x) \leq y \leq g_2(x), \quad a \leq x \leq b$$

$$R: h_1(y) \leq x \leq h_2(y), \quad c \leq y \leq d.$$

Using **FIGURE 9.12.2(a)**, we have

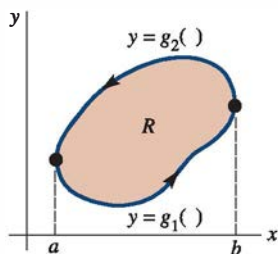
$$\begin{aligned} - \iint_R \frac{\partial P}{\partial y} dA &= - \int_a^b \int_{g_1(x)}^{g_2(x)} \frac{\partial P}{\partial y} dy dx \\ &= - \int_a^b [P(x, g_2(x)) - P(x, g_1(x))] dx \\ &= \int_a^b P(x, g_1(x)) dx + \int_b^a P(x, g_2(x)) dx \\ &= \oint_C P(x, y) dx. \end{aligned} \quad (2)$$

Similarly, from Figure 9.12.2(b),

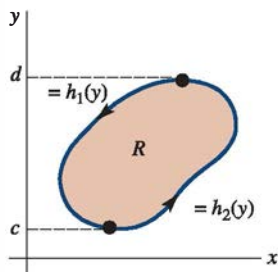
$$\begin{aligned} \iint_R \frac{\partial Q}{\partial x} dA &= \int_c^d \int_{h_1(y)}^{h_2(y)} \frac{\partial Q}{\partial x} dx dy \\ &= \int_c^d [Q(h_2(y), y) - Q(h_1(y), y)] dy \\ &= \int_c^d Q(h_2(y), y) dy + \int_d^c Q(h_1(y), y) dy \\ &= \oint_C Q(x, y) dy. \end{aligned} \quad (3)$$

Adding the results in (2) and (3) yields (1). ≡

The result given in (1) is named after **George Green** (1793–1841), an English mathematician and physicist. The words *in the plane* suggest that the theorem generalizes to 3-space. It does—read on.



(a) R as a Type I region



(b) R as a Type II region

FIGURE 9.12.2 Region R in Theorem 9.12.1

Although the foregoing proof is not valid, the theorem is applicable to more complicated regions, such as those shown in **FIGURE 9.12.3**. The proof consists of decomposing R into a finite number of subregions to which (1) can be applied and then adding the results.

EXAMPLE 1 Using Green's Theorem

Evaluate $\oint_C (x^2 - y^2) dx + (2y - x) dy$, where C consists of the boundary of the region in the first quadrant that is bounded by the graphs of $y = x^2$ and $y = x^3$.

SOLUTION If $P(x, y) = x^2 - y^2$ and $Q(x, y) = 2y - x$, then $\partial P/\partial y = -2y$ and $\partial Q/\partial x = -1$. From (1) and **FIGURE 9.12.4**, we have

$$\begin{aligned}\oint_C (x^2 - y^2) dx + (2y - x) dy &= \iint_R (-1 - 2y) dA \\ &= \int_0^1 \int_{x^3}^{x^2} (-1 - 2y) dy dx \\ &= \int_0^1 \left[-y - y^2 \right]_{x^3}^{x^2} dx \\ &= \int_0^1 (-x^6 + x^4 + x^3 - x^2) dx = -\frac{11}{420}.\end{aligned}$$

We note that the line integral in Example 1 could have been evaluated in a straightforward manner using the variable x as a parameter. However, in the next example, ponder the problem of evaluating the given line integral in the usual manner.

EXAMPLE 2 Using Green's Theorem

Evaluate $\oint_C (x^5 + 3y) dx + (2x - e^{y^3}) dy$, where C is the circle $(x - 1)^2 + (y - 5)^2 = 4$.

SOLUTION Identifying $P(x, y) = x^5 + 3y$ and $Q(x, y) = 2x - e^{y^3}$, we have $\partial P/\partial y = 3$ and $\partial Q/\partial x = 2$. Hence, (1) gives

$$\oint_C (x^5 + 3y) dx + (2x - e^{y^3}) dy = \iint_R (2 - 3) dA = -\iint_R dA.$$

Now the double integral $\iint_R dA$ gives the area of the region R bounded by the circle of radius 2 shown in **FIGURE 9.12.5**. Since the area of the circle is $\pi 2^2 = 4\pi$, it follows that

$$\oint_C (x^5 + 3y) dx + (2x - e^{y^3}) dy = -4\pi.$$

EXAMPLE 3 Work Done by a Force

Find the work done by the force $\mathbf{F} = (-16y + \sin x^2)\mathbf{i} + (4e^y + 3x^2)\mathbf{j}$ acting along the simple closed curve C shown in **FIGURE 9.12.6**.

SOLUTION From (12) of Section 9.8 the work done by $\mathbf{F}$ is given by

$$W = \oint_C \mathbf{F} \cdot d\mathbf{r} = \oint_C (-16y + \sin x^2) dx + (4e^y + 3x^2) dy$$

and so by Green's theorem $W = \iint_R (6x + 16) dA$. In view of the region R , the last integral is best handled in polar coordinates. Since R is defined by $0 \leq r \leq 1$, $\pi/4 \leq \theta \leq 3\pi/4$,

$$W = \int_{\pi/4}^{3\pi/4} \int_0^1 (6r \cos \theta + 16) r dr d\theta$$

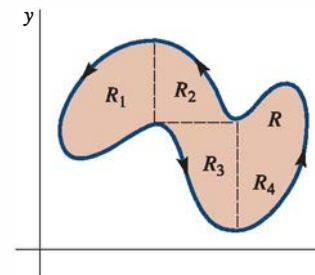


FIGURE 9.12.3 Subregions of R

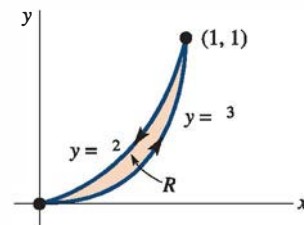


FIGURE 9.12.4 Curve C in Example 1

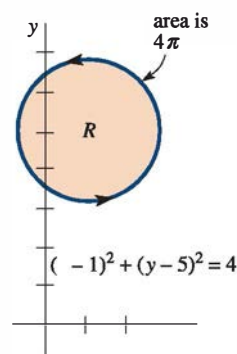


FIGURE 9.12.5 Circular curve C in Example 2

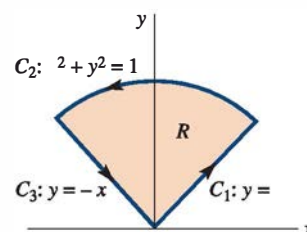


FIGURE 9.12.6 Curve C in Example 3

$$\int_{\pi/4}^{3\pi/4} (2r^3 \cos \theta + 8r^2) \Big|_0^1 d\theta$$

$$\int_{\pi/4}^{3\pi/4} (2 \cos \theta + 8) d\theta = 4\pi.$$

≡

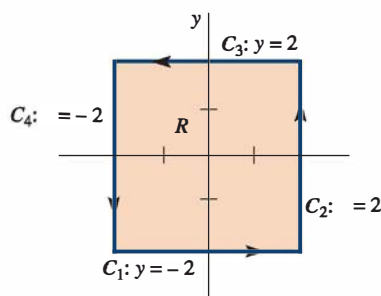
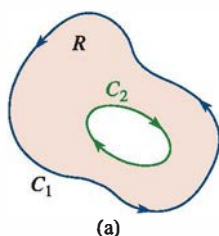
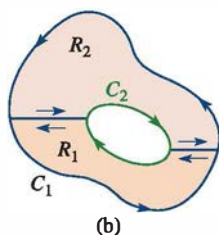


FIGURE 9.12.7 Curve C in Example 4



(a)



(b)

FIGURE 9.12.8 Boundary of R is $C = C_1 \cup C_2$

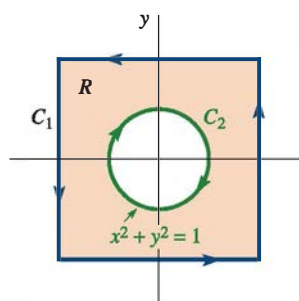


FIGURE 9.12.9 Boundary C in Example 5

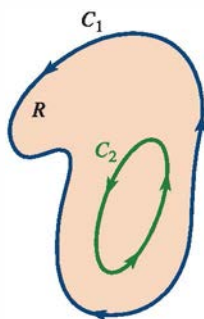


FIGURE 9.12.10 Curves C_1 and C_2 in (5)

EXAMPLE 4 Green's Theorem Not Applicable

Let C be the closed curve consisting of the four straight line segments C_1, C_2, C_3, C_4 shown in FIGURE 9.12.7. Green's theorem is *not* applicable to the line integral

$$\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$$

since $P, Q, \partial P/\partial y$, and $\partial Q/\partial x$ are not continuous at the origin.

≡

Region with Holes Green's theorem can also be extended to a region R with "holes;" that is, a region bounded between two or more piecewise-smooth simple closed curves. In FIGURE 9.12.8(a) we have shown a region R bounded by a curve C that consists of two simple closed curves C_1 and C_2 ; that is, $C = C_1 \cup C_2$. The curve C is positively oriented, since if we traverse C_1 in a counterclockwise direction and C_2 in a clockwise direction, the region R is always to the left. If we now introduce crosscuts as shown in Figure 9.12.8(b), the region R is divided into two subregions, R_1 and R_2 . By applying Green's theorem to R_1 and R_2 , we obtain

$$\iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA = \iint_{R_1} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA + \iint_{R_2} \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

$$= \oint_{C_1} P dx + Q dy + \oint_{C_2} P dx + Q dy$$

$$= \oint_C P dx + Q dy.$$

The last result follows from the fact that the line integrals on the crosscuts (paths with opposite orientations) cancel each other. See (8) of Section 9.8.

EXAMPLE 5 Region with a Hole

Evaluate $\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy$, where $C = C_1 \cup C_2$ is the boundary of the shaded region R shown in FIGURE 9.12.9.

SOLUTION Because $P(x, y) = \frac{-y}{x^2 + y^2}$, $Q(x, y) = \frac{x}{x^2 + y^2}$, and the partial derivatives

$$\frac{\partial P}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2}, \quad \frac{\partial Q}{\partial x} = \frac{y^2 - x^2}{(x^2 + y^2)^2}$$

are continuous on the region R bounded by C , it follows from (4) that

$$\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy = \iint_R \left[\frac{y^2 - x^2}{(x^2 + y^2)^2} - \frac{y^2 - x^2}{(x^2 + y^2)^2} \right] dA = 0.$$

≡

As a consequence of the discussion preceding Example 5 we can establish a result for line integrals that enables us, under certain circumstances, to replace a complicated closed path with a path that is simpler. Suppose, as shown in FIGURE 9.12.10, that C_1 and C_2 are two nonintersecting piecewise-smooth simple closed paths that have the same counterclockwise orientation. Suppose

further that P and Q have continuous first partial derivatives such that

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

in the region R bounded between C_1 and C_2 . Then from (4) above and (8) of Section 9.8 we have

$$\oint_{C_1} P dx + Q dy + \oint_{-C_2} P dx + Q dy = 0$$

or

$$\oint_{C_1} P dx + Q dy = \oint_{C_2} P dx + Q dy. \quad (5)$$

EXAMPLE 6 Example 4 Revisited

Evaluate the line integral in Example 4.

SOLUTION One method of evaluating the line integral is to write

$$\oint_C \int_{C_1} + \int_{C_2} + \int_{C_3} + \int_{C_4}$$

and then evaluate the four integrals on the line segments C_1 , C_2 , C_3 , and C_4 . Alternatively, if we note that the circle $C': x^2 + y^2 = 1$ lies entirely within C (see **FIGURE 9.12.11**), then from Example 5 it is apparent that $P = -y/(x^2 + y^2)$ and $Q = x/(x^2 + y^2)$ have continuous first partial derivatives in the region R bounded between C and C' . Moreover,

$$\frac{\partial P}{\partial y} = \frac{y^2 - x^2}{(x^2 + y^2)^2} = \frac{\partial Q}{\partial x}$$

in R . Hence, it follows from (5) that

$$\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy = \oint_{C'} \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy.$$

Using the parameterization $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$ for C' we obtain

$$\begin{aligned} \oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy &= \int_0^{2\pi} [-\sin t(-\sin t) + \cos t(\cos t)] dt \\ &= \int_0^{2\pi} (\sin^2 t + \cos^2 t) dt \\ &= \int_0^{2\pi} 1 dt = 2\pi. \end{aligned} \quad (6)$$

It is interesting to note that the result in (6):

$$\oint_C \frac{-y}{x^2 + y^2} dx + \frac{x}{x^2 + y^2} dy = 2\pi$$

is true for every piecewise-smooth simple closed curve C with the origin in its interior. We need only choose C' to be $x^2 + y^2 = a^2$, where a is small enough so that the circle lies entirely within C .

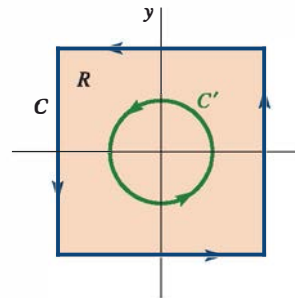


FIGURE 9.12.11 Curves C and C' in Example 6

9.12 Exercises

Answers to selected odd-numbered problems begin on page ANS-23.

In Problems 1–4, verify Green's theorem by evaluating both integrals.

1. $\oint_C (x - y) dx + xy dy = \iint_R (y + 1) dA$, where C is the triangle with vertices $(0, 0)$, $(1, 0)$, $(1, 3)$

2. $\oint_C 3x^2y dx + (x^2 - 5y) dy = \iint_R (2x - 3x^2) dA$, where C is the rectangle with vertices $(-1, 0)$, $(1, 0)$, $(1, 1)$, $(-1, 1)$
3. $\oint_C -y^2 dx + x^2 dy = \iint_R (2x + 2y) dA$, where C is the circle $x = 3 \cos t$, $y = 3 \sin t$, $0 \leq t \leq 2\pi$